Midterm—Algebra
Fall 2024

Algebra Team (2025)

1. A square matrix which cannot be written as the product of elementary matrices is:

(a) rectangular (b) singular (c) nonsingular (d) elementary (e) none

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Consider $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$

2. The elementary row operation performed on a 3×2 matrix when it is multiplied from the left by E:

(a) $3r_3 \to r_3$ (b) $r_2 - 3r_3 \to r_2$ (c) $r_2 + 3r_3 \to r_2$ (d) $-3r_3 \to r_3$ (e) none

3. The entry in row 3 and column 3 in E^{-1} is:

(a) -3 (b) 3 (c) $-1/3$ (d) $1/3$ (e) none

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4. Consider the linear system $A\mathbf{x} = \mathbf{b}$, if the size of A is $n \times n$ and $\mathbf{b} \neq \mathbf{0}$, then the system cannot have:

(a) the trivial solution (b) no solution
(c) infinitely many solutions (d) none

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5. If the matrix $\begin{bmatrix} 1 & 3 & f^2-4 & 2 & 0 & 4 \\ 0 & 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & f-2 & 2 \end{bmatrix}$ is in REF, then f is equal to:

(a) -1 (b) 2 (c) 3 (d) -2 (e) none

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6. If A is an $n \times n$ skew-symmetric nonsingular matrix, then $(A^{-1})^5$ is:

(a) diagonal (b) symmetric (c) triangular
(d) skew-symmetric (e) none

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7. If A, B, C, D and F are $n \times n$ nonsingular matrices, and $ADXCB = F$, then X is equal to:

(a) $D^{-1}A^{-1}FC^{-1}B^{-1}$ (b) $D^{-1}A^{-1}FB^{-1}C^{-1}$ (c) $A^{-1}D^{-1}FC^{-1}B^{-1}$
(d) $A^{-1}B^{-1}FB^{-1}C^{-1}$ (e) none

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8. D, L and U are $n \times n$ nonsingular matrices. D is diagonal, L is lower triangular and U is upper triangular. $(D^2L)^{-1}U^T$ is:

(a) lower triangular (b) upper triangular (c) symmetric
(d) diagonal (e) none

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9. If $(A^T - 0.3I)^{-1} = \begin{bmatrix} 1 & -4 \\ 2 & g \end{bmatrix}$, $g \neq -8$, then the entry in row 1 and column 2 of A is equal to:

(a) $2/(g+8)$ (b) $4/(g+8)$ (c) $-2/(g+8)$ (d) $-4/(g+8)$ (e) none

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Consider the linear system $\begin{cases} x + y + 3z = 2 \\ 4x + 5y + 2z = -2 \\ 3x + y + 29z = 26 \end{cases}$. If $z = t$, $t \in \mathbb{R}$, then:

10. y is equal to:

(a) $-10 - 10t$ (b) $10 + 11t$ (c) $10 - 11t$ (d) $-10 + 10t$ (e) none

11. x is equal to:

(a) $12 - 13t$ (b) $12 + 13t$ (c) $13 + 12t$ (d) $13 - 12t$ (e) none

12. If $A = \begin{bmatrix} 1 & -2 \\ r & 2 \end{bmatrix}$, $r \in \mathbb{R}$, then the entry in row 2 and column 1 of $A^T A^2$ is equal to:

(a) $12r - 12$ (b) $10r - 2$ (c) $15r - 3$ (d) $7r - 8$ (e) none

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Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & -6 \\ -6 & -2 & k \end{bmatrix}$, where $|k| < 8$:

13. The entry in row 2 and column 3 of U is equal to:

(a) 3 (b) 5 (c) 4 (d) -1 (e) none

14. The entry in row 3 and column 3 of L is equal to:

(a) $1/(k+5)$ (b) $1/(k+4)$ (c) $k+4$ (d) $k+5$ (e) none

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15. The linear system $\begin{cases} x + 2y + 3z = a \\ 3x - y + 4z = b \\ 4x + y + 7z = c \end{cases}$, where a, b, $c \in \mathbb{R}$, will be consistent if:

(a) $c = a + b$ (b) $c = -5a + 3b$ (c) $c = 10a - 2b$
(d) $c = -2a + 2b$ (e) none

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16. Consider the linear system $\begin{cases} x + y - z = a \\ 2x + 5y + 2z = -6 \\ 3x + 2y + z = -4 \end{cases}$. If $x = r$, $y = s$ and $z = 3$ is the solution to this system, where r, s, $a \in \mathbb{R}$, then a is equal to:

(a) 6 (b) 1 (c) -6 (d) -1 (e) none

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17. Consider the LU decomposition of the matrix $\begin{bmatrix} 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 5 \\ -4 & 2 & 4 & 6 \\ 6 & 4 & 1 & 2 \end{bmatrix}$. The entry in row 4 and column 4 of L is equal to:

(a) $1/25$ (b) 25 (c) 26 (d) $1/26$ (e) none

Consider $A = \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 3 & 10 & 5 & 17 & 6 & 22 \\ 4 & 14 & 6 & 25 & 9 & 41 \\ 2 & 5 & 5 & 6 & 2 & -4 \end{bmatrix}$. Let R be the reduced row echelon form of A:

18. The entry in row 2 and column 6 of R is equal to:

(a) -44 (b) -53 (c) -47 (d) -50 (e) none

19. The entry in row 1 and column 6 of R is equal to:

(a) 81 (b) 57 (c) 73 (d) 65 (e) none

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20. If A and B are $n \times n$ nonsingular matrices, then $B^2(A^{-1} + B^{-1})A(A + B)^{-1}A$ is equal to:

(a) AB^2 (b) BA (c) AB (d) BA^2 (e) none

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21. The linear system $\begin{cases} 2x + ay - z = 0 \\ -3x + by + 5z = 0 \\ x + cy - 2z = 0 \end{cases}$, where a, b, $c \in \mathbb{R}$, will have nontrivial solutions if:

(a) $4a + 3b + c = 0$ (b) $a + b + c = 0$ (c) $2a + 3b + 5c = 0$
(d) $a + 3b + 7c = 0$ (e) none

Consider the linear system $\begin{bmatrix} 1 & -2 & 4 \\ 2 & -3 & 11 \\ 3 & -4 & 18 \\ 4 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$, where a, b, c, $d \in \mathbb{R}$:

22. The system will be consistent if:

(a) $c = 2b - a$ (b) $c = 4b - 5a$ (c) $c = 3b - 3a$
(d) $c = 5b - 7a$ (e) none

23. The system has a unique solution and $z = 1$, then:

(a) $d = 9b - 14a - 40$ (b) $11b - 18a - 47$ (c) $d = 12b - 20a - 51$
(d) $d = 9b - 14a - 39$ (e) none

Midterm—Algebra
Fall 2024

Algebra Team (2025)

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1. A square matrix which cannot be written as the product of elementary matrices is:

Answer: (b) singular;
if A is reducible to RREF, then $E_k E_{k-1} \dots E_1 A = I$, which means:

$$A = E_1^{-1} E_2^{-1} \dots E_k^{-1} I$$

which means A can be expressed as the product of elementary matrices. So, A has an inverse (is invertible)

$$A^{-1} = E_k E_{k-1} \dots E_1$$

but if a matrix cannot be expressed as the product of elementary matrices, then it has no inverse (is singular/noninvertible).

Consider $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$

2. The elementary row operation performed on a 3×2 matrix when it is multiplied from the left by E:

Answer: (d) $-3r_3 \to r_3$; multiplying a 3×2 matrix from the left by E looks like

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} A_{3\times 2}$$

and generally to obtain the elementary matrix from an elementary row operation on some row of another matrix A, we apply the same operation on the identity matrix I.

In this case, the operation applied would be $-3r_3 \to r_3$.

3. The entry in row 3 and column 3 in E^{-1} is:

Answer: (c) $-1/3$; the inverse of an elementary matrix can be obtained by performing the inverse row operation on the identity matrix.

Or, simply, if it's only multiplying a row by a constant (as in this case), take the reciprocal of that number.

4. Consider the linear system $A\mathbf{x} = \mathbf{b}$, if the size of A is $n \times n$ and $\mathbf{b} \neq \mathbf{0}$, then the system cannot have:

Answer: (a) the trivial solution; for our system, we have three possible types of solutions:

i. a unique solution (or the trivial solution if and only if the system is homogeneous).

ii. infinitely many solutions (including the trivial solution if and only if the system is homogeneous).

iii. no solutions

And so, generally, it has to have one of these three types of solutions. Furthermore, if $A\mathbf{x} = \mathbf{0}$, then the system is homogeneous and has the trivial solution (either as a unique solution or as part of infinitely many solutions).

But our system is not homogeneous, $\mathbf{b} \neq \mathbf{0}$ and so, it cannot have the trivial solution.

5. If the matrix $\begin{bmatrix} 1 & 3 & f^2-4 & 2 & 0 & 4 \\ 0 & 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & f-2 & 2 \end{bmatrix}$ is in REF, then f is equal to:

Answer: (c) 3; REF has specific properties:

i. any zero-rows $\begin{bmatrix} 0 & 0 & \dots & 0 \end{bmatrix}$ appear at the very bottom of the matrix.

ii. the first non-zero entry from the left in any row is a 1 (called a leading one.)

iii. In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row (staircase pattern.)

if we apply property (i) to our matrix, we see that

$$\begin{bmatrix} 0 & 0 & 0 & 0 & f-2 & 2 \end{bmatrix}$$

can never be a zero row. Also, if we picked $f = 2$, property (ii) would not be satisfied for

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

because the first entry from the left is not a leading one. So, to make the first entry from the left a leading one, $f - 2 = 1 \Rightarrow f = 3$. If we substitute it in

$$\begin{bmatrix} 1 & 3 & 5 & 2 & 0 & 4 \\ 0 & 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

which satisfies all 3 properties for REF. (verify yourself)

6. If A is an $n \times n$ skew-symmetric nonsingular matrix, then $(A^{-1})^5$

Answer: (d) skew-symmetric; let $(A^{-1})^5 = D$, then

(a) if D is skew-symmetric; $D^T = -D$

(b) if D is symmetric; $D^T = D$

Now, take the transpose of D

$$\begin{aligned} D^T &= ((A^{-1})^5)^T \\ &= ((A^{-1})^T)^5 && ((A^n)^T = (A^T)^n) \\ &= ((A^T)^{-1})^5 && ((A^{-1})^T = (A^T)^{-1}) \\ &= -(A^{-1})^5 = -D && (A^T = -A \text{ and } (rB)^n = r^n B^n) \end{aligned}$$

So that $(A^{-1})^5$ is skew-symmetric.

7. If A, B, C, D and F are $n \times n$ nonsingular matrices, and $ADXCB = F$, then X is equal to:

Answer: (b) $D^{-1}A^{-1}FB^{-1}C^{-1}$; multiply from the left by $(AD)^{-1}$

$$XCB = (AD)^{-1}F = D^{-1}A^{-1}F$$

multiply from the right by $(CB)^{-1}$

$$X = D^{-1}A^{-1}F(CB)^{-1} = D^{-1}A^{-1}FB^{-1}C^{-1}$$

8. D, L and U are $n \times n$ nonsingular matrices. D is diagonal, L is lower triangular and U is upper triangular. $(D^2L)^{-1}U^T$

Answer: (a) lower triangular; since D is diagonal, then D^2 is also diagonal. One way to understand it is to look at the way diagonal matrices multiply

$$\begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} = \begin{bmatrix} d_1^2 & 0 & \dots & 0 \\ 0 & d_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n^2 \end{bmatrix}$$

which can be intuitively understood by only doing the first few rows and columns.
Additionally, D^2L has to be lower triangular. Again, looking at the way they multiply:

$$\begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \begin{bmatrix} l_{11} & 0 & \dots & 0 \\ l_{21} & l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \dots & l_{nn} \end{bmatrix} = \begin{bmatrix} d_1 l_{11} & 0 & \dots & 0 \\ d_2 l_{21} & d_2 l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ d_n l_{n1} & d_n l_{n2} & \dots & d_n l_{nn} \end{bmatrix}$$

then, using the properties of lower triangular matrices, $(D^2L)^{-1}$ is also lower triangular. And since U^T has to be lower triangular (property of transpose of upper triangular matrices), then $(D^2L)^{-1}U^T$ is also lower triangular.

9. If $(A^T - 0.3I)^{-1} = \begin{bmatrix} 1 & -4 \\ 2 & g \end{bmatrix}$, $g \neq -8$, then the entry in row 1 and column 2 of A is equal to:

Answer: (c) $-2/(g+8)$;

take the inverse of both sides

$$A^T - 0.3I = \begin{bmatrix} 1 & -4 \\ 2 & g \end{bmatrix}^{-1} = \frac{1}{g+8} \begin{bmatrix} g & 4 \\ -2 & 1 \end{bmatrix}$$

add $0.3I$ to both sides

$$A^T = \frac{1}{g+8} \begin{bmatrix} g & 4 \\ -2 & 1 \end{bmatrix} + 0.3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

then, simply take the transpose

$$A = \frac{1}{g+8} \begin{bmatrix} g & -2 \\ 4 & 1 \end{bmatrix} + 0.3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

so the entry in row 1 and column 2 is the sum of the entries in row 1 and column 2 of each matrix $= -2/(g+8)$.

Consider the linear system $\begin{cases} x + y + 3z = 2 \\ 4x + 5y + 2z = -2 \\ 3x + y + 29z = 26 \end{cases}$. If $z = t$, $t \in \mathbb{R}$, then:

10. y is equal to:

Answer: (d) $-10 + 10t$; if we proceed with writing it in augmented matrix form:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 4 & 5 & 2 & -2 \\ 3 & 1 & 29 & 26 \end{array}\right]$$

we do Gaussian elimination (or Gauss-Jordan if you would like to)

$$\xrightarrow[r_3 - 3r_1 \to r_3]{r_2 - 4r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & -10 & -10 \\ 0 & -2 & 20 & 20 \end{array}\right]$$

$$\xrightarrow{r_3 + 2r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & -10 & -10 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

then, let $z = t$, we have $y - 10t = -10 \Rightarrow y = -10 + 10t$.

11. x is equal to:

Answer: (a) $12 - 13t$; from the previous problem

$$x + y + 3z = 2 \Rightarrow x = 2 - (y + 3z)$$

then, substitution of y and z

$$x = 2 - (-10 + 10t + 3t) = 12 - 13t$$

12. If $A = \begin{bmatrix} 1 & -2 \\ r & 2 \end{bmatrix}$, $r \in \mathbb{R}$, then the entry in row 2 and column 1 of $A^T A^2$ is equal to:

Answer: (b) $10r - 2$; firstly, A^T

$$A^T = \begin{bmatrix} 1 & r \\ -2 & 2 \end{bmatrix}$$

and $A^2 = AA$

$$A^2 = \begin{bmatrix} 1 & -2 \\ r & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ r & 2 \end{bmatrix} = \begin{bmatrix} 1-2r & -6 \\ 3r & -2r+4 \end{bmatrix}$$

then the multiplication can be carried out easily

$$A^T A^2 = \begin{bmatrix} 1 & r \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 1-2r & -6 \\ 3r & -2r+4 \end{bmatrix}$$

to speed things up, we notice that they're asking for the entry in **row 2** and **column 1**. Then we multiply row 2 of the first matrix by column 1 of the second

$$-2 \times (1-2r) + 2 \times 3r = -2 + 4r + 6r = 10r - 2$$

Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & -6 \\ -6 & -2 & k \end{bmatrix}$, where $|k| < 8$

13. The entry in row 2 and column 3 of U is equal to:

Answer: (c) 4; we use the book-keeping method as follows:

$$\xrightarrow{(1/2)r_1 \to r_1} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 4 & -1 & -6 \\ -6 & -2 & k \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ & & 0 \\ & & \end{bmatrix}$$

$$\xrightarrow[r_3+6r_1 \to r_3]{r_2-4r_1 \to r_2} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 0 & -3 & -12 \\ 0 & 1 & k+9 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 4 & & 0 \\ -6 & & \end{bmatrix}$$

$$\xrightarrow{(-1/3)r_2 \to r_2} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 0 & 1 & 4 \\ 0 & 1 & k+9 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 4 & -3 & 0 \\ -6 & & \end{bmatrix}$$

$$\xrightarrow{r_3-r_2 \to r_3} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 0 & 1 & 4 \\ 0 & 0 & k+5 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 4 & -3 & 0 \\ -6 & 1 & \end{bmatrix}$$

$$\xrightarrow{(1/(k+5))r_3 \to r_3} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 4 & -3 & 0 \\ -6 & 1 & k+5 \end{bmatrix}$$

so that finally, the entry in row 2 and column 3 of U is 4.

14. The entry in row 3 and column 3 of L is equal to

Answer: (d) $k+5$; from the previous problem

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 4 & -3 & 0 \\ -6 & 1 & k+5 \end{bmatrix}$$

and so the entry in row 3 and column 3 is $k+5$.

15. The linear system $\begin{cases} x+2y+3z=a \\ 3x-y+4z=b \\ 4x+y+7z=c \end{cases}$, where a, b, $c \in \mathbb{R}$, will be consistent if:

Answer: (a) $c=a+b$; writing down the system in augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 3 & -1 & 4 & b \\ 4 & 1 & 7 & c \end{array}\right]$$

then using row reduction

$$\xrightarrow[r_3-4r_1 \to r_3]{r_2-3r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & -7 & -5 & b-3a \\ 0 & -7 & -5 & c-4a \end{array}\right]$$

$$\xrightarrow{r_3-r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & -7 & -5 & b-3a \\ 0 & 0 & 0 & c-b-a \end{array}\right]$$

the system is consistent if the final row is $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \end{array}\right]$ corrosponding to infinitely many solutions. Then, $c-b-a=0 \Rightarrow c=a+b$.

16. Consider the linear system $\begin{cases} x+y-z=a \\ 2x+5y+2z=-6 \\ 3x+2y+z=-4 \end{cases}$. If $x=r$, $y=s$ and $z=3$ is the solution to this system, where r, s, $a \in \mathbb{R}$, then a is equal to:

Answer: (c) -6; writing down the system in augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 2 & 5 & 2 & -6 \\ 3 & 2 & 1 & -4 \end{array}\right]$$

then using Gaussian elimination

$$\xrightarrow[r_3-3r_1 \to r_3]{r_2-2r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & 3 & 4 & -6-2a \\ 0 & -1 & 4 & -4-3a \end{array}\right]$$

$$\xrightarrow{(1/3)r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & 1 & 4/3 & -2-2a/3 \\ 0 & -1 & 4 & -4-3a \end{array}\right]$$

$$\xrightarrow{r_3+r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & 1 & 4/3 & -2-2a/3 \\ 0 & 0 & 16/3 & -6-11a/3 \end{array}\right]$$

we could stop here and take that final row,

$$0x + 0y + 16z/3 = -6 - 11a/3 \Rightarrow 16 + 6 = -11a/3 \Rightarrow a = -6$$

17. Consider the LU decomposition of the matrix $\begin{bmatrix} 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 5 \\ -4 & 2 & 4 & 6 \\ 6 & 4 & 1 & 2 \end{bmatrix}$. The entry in row 4 and column 4 of L is equal to:

Answer: (b) 25; using the book-keeping method

$$\xrightarrow{(1/4)r_1 \to r_1} \begin{bmatrix} 1 & 0 & 1/2 & 3/2 \\ 2 & 1 & 3 & 5 \\ -4 & 2 & 4 & 6 \\ 6 & 4 & 1 & 2 \end{bmatrix} \qquad L = \begin{bmatrix} 4 & 0 & 0 & 0 \\ & & 0 & 0 \\ & & & 0 \\ & & & \end{bmatrix}$$

$$\xrightarrow[r_4-6r_1 \to r_4]{r_2-2r_1 \to r_2,\ r_3+4r_1 \to r_3} \begin{bmatrix} 1 & 0 & 1/2 & 3/2 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 6 & 12 \\ 0 & 4 & -2 & -7 \end{bmatrix} \qquad L = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 2 & & 0 & 0 \\ -4 & & & 0 \\ 6 & & & \end{bmatrix}$$

$$\xrightarrow[r_4-4r_2 \to r_4]{r_3-2r_2 \to r_3} \begin{bmatrix} 1 & 0 & 1/2 & 3/2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & -10 & -15 \end{bmatrix} \qquad L = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -4 & 2 & & 0 \\ 6 & 4 & & \end{bmatrix}$$

$$\xrightarrow{(1/2)r_3 \to r_3} \begin{bmatrix} 1 & 0 & 1/2 & 3/2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -10 & -15 \end{bmatrix} \qquad L = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -4 & 2 & 2 & 0 \\ 6 & 4 & & \end{bmatrix}$$

$$\xrightarrow{r_4+10r_3 \to r_4} \begin{bmatrix} 1 & 0 & 1/2 & 3/2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 25 \end{bmatrix} \qquad L = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -4 & 2 & 2 & 0 \\ 6 & 4 & -10 & \end{bmatrix}$$

$$\xrightarrow{(1/25)r_4 \to r_4} \begin{bmatrix} 1 & 0 & 1/2 & 3/2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -4 & 2 & 2 & 0 \\ 6 & 4 & -10 & 25 \end{bmatrix}$$

so the entry in row 4 and column 4 of L is 25.

Consider $A = \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 3 & 10 & 5 & 17 & 6 & 22 \\ 4 & 14 & 6 & 25 & 9 & 41 \\ 2 & 5 & 5 & 6 & 2 & -4 \end{bmatrix}$. Let R be the reduced row echelon form of A

18. The entry in row 2 and column 6 of R is equal to:

Answer: (a) -44; using Gauss-Jordan elimination to reach RREF

$$\xrightarrow[r_4-2r_1 \to r_4]{r_2-3r_1 \to r_2,\ r_3-4r_1 \to r_3} \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 0 & 1 & -1 & 2 & 3 & 4 \\ 0 & 2 & -2 & 5 & 5 & 17 \\ 0 & -1 & 1 & -4 & 0 & -16 \end{bmatrix}$$

$$\xrightarrow[r_4+r_2 \to r_4]{r_1-3r_2 \to r_1,\ r_3-2r_2 \to r_3} \begin{bmatrix} 1 & 0 & 5 & -1 & -8 & -6 \\ 0 & 1 & -1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & -1 & 9 \\ 0 & 0 & 0 & -2 & 3 & -12 \end{bmatrix}$$

$$\xrightarrow[r_4+2r_3 \to r_4]{r_1+r_3 \to r_1,\ r_2-2r_3 \to r_2} \begin{bmatrix} 1 & 0 & 5 & 0 & -9 & 3 \\ 0 & 1 & -1 & 0 & 5 & -14 \\ 0 & 0 & 0 & 1 & -1 & 9 \\ 0 & 0 & 0 & 0 & 1 & 6 \end{bmatrix}$$

$$\xrightarrow[r_3+r_4 \to r_3]{r_1+9r_4 \to r_1,\ r_2-5r_4 \to r_2} \begin{bmatrix} 1 & 0 & 5 & 0 & 0 & 57 \\ 0 & 1 & -1 & 0 & 0 & -44 \\ 0 & 0 & 0 & 1 & 0 & 15 \\ 0 & 0 & 0 & 0 & 1 & 6 \end{bmatrix}$$

then the entry in row 2 and column 6 of R is -44.

19. The entry in row 1 and column 6 of R is equal to:

Answer: (b) 57; from the previous problem, the entry in row 1 and column 6 is 57.

20. If A and B are $n \times n$ nonsingular matrices, then $B^2(A^{-1} + B^{-1})A(A + B)^{-1}A$ is equal to:

Answer: (b) BA;

$$\begin{aligned} B^2(A^{-1} + B^{-1})A(A + B)^{-1}A &= (B^2A^{-1} + B)A(A + B)^{-1}A \\ &= (B^2 + BA)(A + B)^{-1}A \end{aligned}$$

then factor out a B from the left

$$B^2(A^{-1} + B^{-1})A(A + B)^{-1}A = B\underbrace{(B + A)(A + B)^{-1}}_{I}A = BA$$

21. The linear system $\begin{cases} 2x + ay - z = 0 \\ -3x + by + 5z = 0 \\ x + cy - 2z = 0 \end{cases}$, where a, b, $c \in \mathbb{R}$, will have nontrivial solutions if:

Answer: (d) $a + 3b + 7c = 0$; writing our system in matrix equation form

$$\begin{bmatrix} 2 & a & -1 \\ -3 & b & 5 \\ 1 & c & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

since our system is homogeneous, the only possible types of solutions are:

i. a unique solution (the trivial solution).

ii. infinitely many solutions (which include the trivial solution **and** the nontrivial solutions).

then we're looking for infinitely many solutions. Since our coefficient matrix is square, we could calculate the determinant and determine the type of solution from it.

If the type is infinitely many solutions, then the determinant has to be 0. (calculating using the middle column)

$$\begin{vmatrix} 2 & a & -1 \\ -3 & b & 5 \\ 1 & c & -2 \end{vmatrix} = -a \times (6 - 5) + b \times (-4 + 1) - c \times (10 - 3) = -a - 3b - 7c = 0$$

then $a + 3b + 7c = 0$.

Consider the linear system $\begin{bmatrix} 1 & -2 & 4 \\ 2 & -3 & 11 \\ 3 & -4 & 18 \\ 4 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$, where a, b, c, $d \in \mathbb{R}$

22. The system will be consistent if:

Answer: (a) $c = 2b - a$; putting our system into augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 2 & -3 & 11 & b \\ 3 & -4 & 18 & c \\ 4 & 4 & 1 & d \end{array}\right]$$

and using row reduction

$$\xrightarrow[r_4 - 4r_1 \to r_4]{r_2 - 2r_1 \to r_2,\ r_3 - 3r_1 \to r_3} \left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b - 2a \\ 0 & 2 & 6 & c - 3a \\ 0 & 12 & -15 & d - 4a \end{array}\right]$$

$$\xrightarrow[r_4 - 12r_2 \to r_4]{r_3 - 2r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b - 2a \\ 0 & 0 & 0 & c - 2b + a \\ 0 & 0 & -51 & d - 12b + 20a \end{array}\right]$$

$$\xrightarrow{r_3 \leftrightarrow r_4} \left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b-2a \\ 0 & 0 & -51 & d-12b+20a \\ 0 & 0 & 0 & c-2b+a \end{array}\right]$$

we could stop here and start our solution. It's apparent that the system is consistent if:

$$\#\text{leading variables} = \#\text{non-zero rows} = \#\text{unknowns} = 3$$

that only happens if $c - 2b + a = 0 \Rightarrow c = 2b - a$.

23. The system has a unique solution and $z = 1$, then:

Answer: **(c)** $d = 12b - 20a - 51$; from the previous problem, the third row expresses the equation

$$0x + 0y - 51z = d - 12b + 20a$$

then, simply substitute $z = 1$ and rearrange

$$d = 12b - 20a - 51$$